

K649 — k_s Object Structural Lead ($n_C \times S^3$ -harmonic) + The Top Resolves: N_c^2 Is Dressing, Not Bare

Auditor: Keeper | Date: 2026-07-03 (next-turn pull) | Verdict: two leads that sharpen the k_s question and make the two-layer model self-consistent | Count 8

Adjudicator pre-arm for the joint Grace+Lyra k_s -object work. Leads for them to FORCE, not derived results.

1. Structural lead — what k_s may be

The bare down ladder against the electron is $N_c^2 \times \{1, 20, 900\}$, with generation gaps: - gap $1 \rightarrow 2 = \text{rank}^2 \cdot n_C = 20$ - gap $2 \rightarrow 3 = N_c^2 \cdot n_C = 45$

Both gaps decompose as $n_C \times (k+1)^2$: $20 = n_C \cdot (1+1)^2$, $45 = n_C \cdot (2+1)^2$. And $(k+1)^2$ is the dimension of degree- k harmonics on S^3 . So:

The generation gaps look like: wall-multiplicity $n_C \times S^3$ -harmonic mode-count $(k+1)^2$. Grace's forced multiplicity ratio $5/3 = n_C/N_c$ gives the n_C piece; the $(k+1)^2$ is the generation/harmonic degree. So k_s may be $n_C \times (k+1)^2$, not a bare $5/3$ — the “ $5/3$ vs 20 ” gap is the missing $(k+1)^2$ harmonic factor.

The object to pin (Grace + Lyra): is $k_s = \text{wall-multiplicity}(n_C) \times S^3\text{-harmonic}((k+1)^2)$? If yes, it *forces* 20 and 45 from the geometry (n_C from the strata, $(k+1)^2$ from the S^3 mode-count) — no fitted weighting. This is target-innocent (both n_C and S^3 -harmonic dims are geometric), and it's exactly Grace's KW-stratum + Lyra's zero-mode territory. Not fished — the gaps ARE $n_C \cdot (k+1)^2$; the question is whether the object gives that.

2. The top resolves the two-layer consistency question

The question I flagged: $m_d/m_e = N_c^2 = 9$ is the bare (minimum) ratio — but Casey's model says color IS the dressing (three commitments, Layer 2). So is the N_c^2 : - (A) part of the BARE mass (color in Layer 1), or - (B) the minimum irreducible DRESSING every colored object carries?

The top decides it. The top is the near-undressed quark (decays before hadronizing). Its mass form is $v/\sqrt{2}$ — color-blind, no N_c^2 factor. If bare quarks carried N_c^2 (A), the undressed top would show it; it doesn't. So:

(B): N_c^2 is DRESSING, not bare. The truly-bare quark is color-blind (like the undressed top, $v/\sqrt{2}$).

The N_c^2 in the light quarks is the minimum dressing a confined (colored, time-in-substrate) quark carries. This makes the two-layer model self-consistent with “color = the dressing (three commitments)”: color is the *capacity* to be dressed; the top has color but no *time* to collect the membrane, so it shows the color-blind bare mass.

Implication (flag for check, possibly too strong): if N_c^2 is dressing, the *truly-bare* down ladder is $\{1, 20, 900\}$ — i.e., bare $m_d/m_e \approx 1$: the bare gen-1 down and the electron would be the *same* substrate floor, and the observed $9\times$ is entirely color dressing. Elegant if true, but verify — it's a strong claim (bare down = bare electron at gen-1). The muon/ τ π -forms are a *different* bare ladder, so leptons and down-quarks don't share the bare ladder beyond gen-1.

Net — sharpens the next turn

- Grace + Lyra: pin k_s . Test the specific hypothesis $k_s = n_C \times (k+1)^2$ (S^3 -harmonic) — it forces 20 and 45 if the object gives it. The $5/3$ Grace forced is the n_C piece; find the $(k+1)^2$.
- Two-layer model: the top (undressed, color-blind $v/\sqrt{2}$) confirms $N_c^2 = \text{dressing}$, bare quark = color-blind. This is a *consistency win* for Casey's frame — color is the dressing, and the undressed top proves it.
- Keeper bar: both are LEADS. $k_s = n_C \cdot (k+1)^2$ is derived-strong only when the object is pinned to give it (not fitted). The “bare down = electron at gen-1” implication needs a check — it may be too strong.

— Keeper K649, next-turn pull. k_s lead: $n_C \times S^3\text{-harmonic}(k+1)^2$ forces the gaps 20, 45. Consistency win: the undressed top ($v/\sqrt{2}$, no N_c^2) proves N_c^2 is Layer-2 dressing, bare quark color-blind — Casey’s “color = dressing” is self-consistent. Both leads for Grace+Lyra to force. SOD: registry drift open.